

National Land Acquisition & Management System

Smart India Hackathon (SIH) 2026 Technical Report

MEMBER PROFILE: Member 2

Technical Role: Backend Core & Database Engineer

Primary Focus: FastAPI routing modules & controller logic setup, Pydantic schema validation boundaries implemen

1. Functional Domain & Work Done

As part of the NLAMS development team, Member 2 has taken ownership of the functional module targeting Backend Core & Database Engineer. The following components have been fully implemented and integrated into the live workspace:

- **Parameterized Query Guards:** Ensured all database queries are fully parameterized via SQLAlchemy, resolving potential SQL Injection risks.
- **Write-Access Authentication:** Exposed dependencies to verify API headers (X-NLAMS-API-Key) protecting POST, PUT, and DELETE endpoints.
- **Dynamic Project Seeding:** Executed and verified seed files establishing 2,501 records (41 projects, 1500 parcels, 960 R&R families).

2. Overall Website Layout & Features

The NLAMS portal operates as a single source of truth under a modernised, high-contrast, official government light theme (saffron, navy, and emerald accents). Key features include:

- **Tricolor Home & Directives:** An entry portal showcasing the LARR Act 2013 roadmap and operational pillars.
- **Bilingual Language Switcher:** A global toggle translating all KPI labels, timelines, Gazette notifications, and Treasury advice receipts dynamically into Hindi and English.
- **Executive Dashboard & GIS:** Renders land parcels as Leaflet map overlay polygons, color-coded by acquisition stage, alongside Recharts predictive dashboards.
- **Automated LARR Workflows:** A pipeline allowing Central, State, and District roles to sign off on stages, instantly generating printable official Gazette templates.
- **Web3 Trust & Compensation Portal:** Simulates smart contract escrow releases with Metamask wallet connectivity, a digital deed SHA-256 validation scanner, and a scrolling block ledger.
- **Mobile-Responsive Field Survey Node:** Allows field surveyors to capture coordinate coordinates using GPS triangulation and report soil classification, structures, and site images.
- **SMS Alert Simulator:** Fires dynamic, floating mobile network text notification alerts during proposal submissions, boundary checks, and DBT escrow releases.

3. Sequential Learning Pathway & Project Roadmap

To ensure a smooth transition from prototype to production for the SIH finals, the following learning benchmarks are mapped for this role:

- **Focus Area:** FastAPI routing modules & controller logic setup
- **Focus Area:** Pydantic schema validation boundaries implementation
- **Focus Area:** SQLAlchemy ORM relations design (Projects, Parcels, Beneficiaries)
- **Focus Area:** Database transactional locks to prevent race conditions on inspection handovers

4. Judge Defense Recommendations

During the pitch and technical Q&A, the judges will test the integrity of each module. Use the following defense arguments to demonstrate technical authority:

- **Defense Tip:** Highlight that database transactions use sqlalchemy parameterized queries to prevent SQL injections, and strict XSS sanitization checks are enforced on all text columns.